

ORIGINAL ARTICLE

Enhancing English Reading Comprehension, Learning Motivation and Attitude Through AI-Supported Pre-Reading Scaffolding

Chih-Chung Lin¹  | Tzu-Hsuan Lin² | Chi-Kay Tang²

¹Language Center, National Chiayi University, Chiayi City, Taiwan | ²Department of Management Information Systems, National Chiayi University, Chiayi City, Taiwan

Correspondence: Chih-Chung Lin (chihchunglin09@gmail.com)

Received: 10 June 2025 | **Revised:** 12 August 2025 | **Accepted:** 19 October 2025

Funding: This work was supported by the National Science and Technology Council, MOST 114-2410-H-415-041.

Keywords: English reading performance | generative artificial intelligence (Gen-AI) | learning attitude | learning motivation

ABSTRACT

Background: The integration of generative artificial intelligence (Gen-AI) into second language (L2) education has opened new possibilities for personalized and adaptive instruction. While Gen-AI has shown promise in supporting language production skills, its application in reading, particularly in pre-reading scaffolding, remains underexplored.

Objectives: This study aimed to examine the effects of AI-supported pre-reading scaffolding, using ChatGPT, on university EFL students' English reading comprehension, learning motivation, and attitudes towards English learning, compared to traditional teacher-led scaffolding.

Methods: A quasi-experimental pretest-posttest design was employed with 66 first-year university EFL students. Participants were assigned to either a teacher-led control group ($n = 34$) or an AI-supported experimental group ($n = 32$). Both groups received structured question-posing scaffolding prior to reading tasks, but delivery methods differed. The control group interacted with instructors, while the experimental group used a custom-designed online platform integrating ChatGPT. ANCOVA was used to analyse the post-intervention outcomes across three domains: reading comprehension, motivation, and attitudes.

Results: ANCOVA results indicated that students in the AI-supported group outperformed the teacher-led group in English reading comprehension, $F(1, 63) = 5.06$, $p = 0.02$, partial $\eta^2 = 0.07$. Similarly, learning motivation was higher in the AI group, $F(1, 63) = 5.90$, $p = 0.01$, partial $\eta^2 = 0.08$. For learning attitude, the AI-supported group also demonstrated an advantage, $F(1, 63) = 14.77$, $p < 0.001$, partial $\eta^2 = 0.19$. These results suggest a moderate-to-large effect of AI-supported pre-reading scaffolding on both cognitive and affective outcomes.

Conclusions: Integrating Gen-AI into pre-reading scaffolding enhances both cognitive and affective outcomes in EFL reading contexts. These findings support the potential of AI-based instructional tools to provide scalable, interactive, and learner-centered support for reading development in higher education.

1 | Introduction

English reading comprehension constitutes a pivotal skill in English as a Foreign Language (EFL) higher education,

serving not only as a gateway to disciplinary knowledge but also as a foundation for academic literacy, critical engagement, and long-term language development (McCarron and Kuperman 2022). It involves complex cognitive processes

Summary

- What is already known on this topic
 - Pre-reading scaffolding is widely used to support English reading comprehension, particularly in EFL contexts.
 - Generative AI tools such as ChatGPT have shown potential in enhancing writing and speaking skills in language learning.
 - Most existing applications of Gen-AI in L2 learning focus on production-oriented tasks; its role in comprehension-based instruction remains underexplored.
 - Learners' reading comprehension is significantly affected by affective factors, including motivation and attitude.
- What this paper adds
 - Provides empirical evidence that AI-supported pre-reading scaffolding (using ChatGPT) improves university students' English reading comprehension compared to traditional teacher-led scaffolding.
 - Demonstrates that AI-assisted scaffolding not only enhances cognitive outcomes but also significantly improves learners' motivation and attitudes towards English.
 - Introduces a structured question-posing AI system (AIPQP) that operationalizes procedural scaffolding through Gen-AI interaction.
 - Validates the feasibility of integrating generative AI tools into pre-reading phases without compromising instructional consistency.
- Implications for practice and/or policy
 - EFL instructors can adopt AI-based scaffolding as a flexible, scalable alternative to traditional pre-reading instruction, especially in large or resource-limited classes.
 - Teacher training programs should include AI prompt engineering and dialogic scaffolding design to align AI tools with pedagogical goals.
 - Educational institutions may integrate Gen-AI tools into blended or online language-learning platforms to enhance learner autonomy and engagement.
 - Policymakers and curriculum developers should consider embedding AI-supported scaffolding frameworks into official English reading curricula to support both cognitive and affective learner development.

such as lexical access, syntactic parsing, inference generation, and the integration of textual information with prior knowledge (Rumelhart and McClelland 2017). Importantly, the effectiveness of reading comprehension is mediated by affective variables; notably, learning motivation, defined as the willingness to sustain effort and engagement in reading tasks (Dörnyei 2001), and learning attitudes, encompassing learners' evaluative beliefs and emotional orientations towards English learning (Li and Gan 2022). Empirical studies underscore the reciprocal relationship among these constructs: higher motivation and positive attitudes facilitate the use of metacognitive and cognitive strategies, while successful

comprehension experiences reinforce motivational and attitudinal dispositions (Huang and Hong 2016).

While traditional teacher-led pre-reading activities, such as guided questioning and vocabulary previews, are pedagogically valuable (Clark and Graves 2005), their effectiveness in university EFL contexts is often constrained by structural limitations. Large class sizes, fixed lesson times, and heterogeneous learner profiles make it challenging to deliver the individualized, differentiated support necessary for effective scaffolding (Bahari et al. 2024; Wijaya 2023). Consequently, opportunities for immediate feedback and iterative meaning negotiation are limited, often resulting in generic guidance that fails to address specific cognitive or affective barriers for every student. It is this gap between the need for personalized scaffolding and the practical constraints of traditional instruction that generative AI is uniquely positioned to address.

Recent advances in generative artificial intelligence (Gen-AI), particularly large language models (LLMs) such as ChatGPT, offer a transformative opportunity to overcome these constraints. AI-supported scaffolding can be defined as the deployment of LLM-driven interactive systems to deliver adaptive, context-sensitive, and feedback-rich support during the pre-reading stage (Kasneci et al. 2023). Theoretically, such systems align with scaffolding theory by providing graduated assistance, prompting learners to generate predictions, identify key ideas, and engage in inferential reasoning, while dynamically adjusting to individual responses (Caines et al. 2023). Empirically, AI-mediated reading interventions have demonstrated potential to enhance both comprehension and affective engagement by offering immediate clarification, recursive elaboration, and a psychologically safe space for inquiry (Murphy Odo 2025; Waluyo and Kusumastuti 2024). Furthermore, by reducing cognitive load through structured prompts and personalised feedback, AI-supported scaffolding may foster deeper processing and more favourable attitudes towards English learning (Gruhn et al. 2024).

Despite the growing body of research on generative AI in language education following the release of ChatGPT in late 2022 (e.g., Crompton et al. 2024; Jawaid et al. 2025; Kristiawan et al. 2024), most empirical studies have focused on speaking, writing, or general language learning support. Only a few studies have investigated AI-mediated reading instruction (e.g., Chea and Xiao 2024; Tang and Foley 2022), and these have primarily emphasised while-reading comprehension checks or post-reading summarization. Comparative evidence on AI's role in the pre-reading phase where scaffolding is most crucial for activating prior knowledge, orienting attention, and fostering motivation is virtually absent. Furthermore, no studies to date have directly contrasted AI-supported pre-reading scaffolding, such as using an LLM to generate personalised questions and vocabulary lists, with traditional teacher-led approaches in university EFL contexts, nor examined their differential impacts on cognitive (reading comprehension) and affective (motivation, attitudes) outcomes.

Thus, to address the mentioned research gap, the present study employs a mixed-methods approach to systematically compare these two approaches to provide both empirical evidence and

pedagogical insights into the integration of generative AI in EFL reading instruction. Specific research questions are addressed as follows:

RQ1. *Do students using AI-integrated pre-reading scaffolding demonstrate better reading comprehension than those receiving teacher-led scaffolding?*

RQ2. *Do students using AI-integrated pre-reading scaffolding demonstrate better English learning motivation than those receiving teacher-led scaffolding?*

RQ3. *Do students using AI-integrated pre-reading scaffolding hold more positive attitudes towards English learning than those receiving teacher-led scaffolding?*

RQ4. *What are the students' perception of using AI-supported pre-reading scaffolding?*

2 | Literature Review

2.1 | Scaffolding Pre-Reading Activities in EFL Contexts

Within EFL contexts, English reading is widely acknowledged as a foundational literacy skill that underpins both academic achievement and lifelong learning trajectories. It serves as a primary conduit for accessing authentic linguistic input, acquiring discipline-specific knowledge, and engaging with culturally embedded discourse conventions, thereby fostering learners' linguistic competence and intercultural communicative awareness (Taye and Teshome 2025). Beyond vocabulary acquisition and grammatical awareness, proficient reading fosters higher-order cognitive skills, such as inference-making, critical evaluation, and integration of multiple information sources (Fitria & Nadiyah, Fitria and Nafiah 2025). Recent studies reaffirm that reading comprehension correlates strongly with overall language proficiency and performance in other skills, including writing and speaking (Ye and Liu 2023). Consequently, enhancing learners' reading competence remains a central pedagogical goal in EFL instruction.

To achieve this goal, instructional models often adopt a staged approach to reading, with pre-reading activities positioned as an essential preparatory phase for optimising comprehension and engagement. Pre-reading scaffolding refers to strategically designed instructional interventions that prepare learners cognitively, linguistically, and affectively for engagement with a text (Graves and Graves 2003). These activities aim to activate relevant schemata, stimulate curiosity, set reading purposes, and reduce potential comprehension barriers (Cho and Ma 2020). Common approaches include charts, anticipation guides, semantic mapping, and guided questioning (Cho and Ma 2020). In EFL classrooms, pre-reading scaffolding has been shown to enhance not only comprehension but also reading motivation and self-efficacy (Li and Gan 2022). However, such scaffolding is predominantly implemented in a teacher-led, whole-class format, which, although beneficial in providing structured support, often constrains personalization due to large class sizes, time limitations, and varying learner profiles.

The pedagogical effectiveness of these pre-reading activities is theoretically grounded in Vygotsky's Sociocultural Theory (SCT), which posits that cognitive development is mediated through social interaction and the use of cultural tools (Vygotsky 1978). Central to SCT is the concept of the Zone of Proximal Development (ZPD), defined as the distance between learners' independent performance and their potential performance with appropriate guidance (Wood et al. 1976). Scaffolding, in this context, functions as a dynamic and adaptive form of assistance that enables learners to accomplish tasks beyond their current capacities, progressively internalizing the strategies and knowledge required for autonomous performance (Masava et al. 2023). Within pre-reading instruction, scaffolding bridges learners' prior knowledge and the demands of the upcoming text, thereby facilitating comprehension and promoting both cognitive and affective engagement (Suryani et al. 2023). This theoretical grounding provides a robust explanatory basis for the pedagogical value of pre-reading scaffolding in EFL contexts, as it elucidates the mechanisms through which scaffolded support can mediate learners' cognitive and affective engagement, ultimately contributing to measurable improvements in reading comprehension and related learning outcomes.

To determine if such improvements have occurred, a clear framework for evaluating reading comprehension is necessary. For assessment purposes, English reading comprehension is recognised as a multidimensional construct encompassing distinct but interrelated levels of cognitive engagement with text. Informed by prior models of reading processes (Tschense and Wallot 2022), L2 reading proficiency assessments often incorporate items targeting multiple layers of comprehension, from literal understanding of text details, through inferential meaning-making, to evaluative engagement, while also reflecting strategic (metacognitive) processes that support comprehension. Recent studies affirm this multidimensional and hierarchical nature of reading comprehension, highlighting differential contributions from literal, inferential, critical/evaluative, and metacognitive components (de-la-Peña and Luque-Rojas 2021; Vettori et al. 2024). This hierarchical framework, progressing from literal through strategic comprehension, not only offers a theoretical basis for understanding the cognitive demands of reading but also provides a principled foundation for designing L2 reading comprehension assessments. In the present study, this framework directly informed the development of the English reading comprehension tests, ensuring that items systematically represent all four comprehension levels and align with the cognitive and metacognitive processes targeted by the AI-supported pre-reading scaffolding intervention.

Based on this multidimensional perspective of reading comprehension, recent advances in educational technology have opened new avenues for tailoring instruction and assessment to these varied cognitive demands. Generative AI (Gen-AI) tools hold the potential to deliver targeted scaffolding aligned with specific comprehension levels, ranging from literal recall to strategic, metacognitive monitoring, while adapting to individual learner needs in real time. Such affordances position Gen-AI as a promising complement to established pedagogical frameworks, warranting closer examination of its role in supporting English reading development.

2.2 | Gen-AI and Its Potential for Supporting English Reading

Recent research underscores a critical distinction in the application of Gen-AI for language learning: its effectiveness is contingent not on the technology itself, but on its pedagogical integration. A case in point is the contrast between studies on simple text simplification and those embedding AI within a structured scaffolding framework. For instance, Murphy Odo (2025) found no significant difference in reading comprehension when learners used AI-generated versus human-simplified texts. This null result suggests that deploying AI atheoretically for tasks like text simplification may be insufficient to produce measurable gains, leading the author to question the tool's value when used without a principled pedagogical framework. Conversely, studies that integrate Gen-AI more deliberately show greater promise. Çelik et al. (2024), for example, demonstrated that when texts simplified by ChatGPT were part of a structured activity, learners' comprehension and inference performance significantly improved. This contrast reinforces the premise of the present study: the affordances of AI must be deliberately aligned with established scaffolding methodologies to optimize their educational impact.

This principle of pedagogical alignment is echoed across various emerging applications of Gen-AI. Research highlights the potential for engineering LLMs as dialogic scaffolding agents (Caines et al. 2023), which can provide the dynamic, personalised feedback crucial for shaping learners' comprehension processes (Gruhn et al. 2024). The educational benefits are further amplified in multi-modal AI systems that integrate textual, visual, and dialogic supports to enhance both comprehension and learner motivation (Hamid 2025). Moreover, the capacity of models like GPT-3 to generate reading passages with controlled difficulty and calibrated Lexile levels offers new possibilities for truly adaptive reading design (Bezirhan and von Davier 2023). Collectively, these findings illustrate that Gen-AI's value lies in its deployment as a responsive, adaptive, and multi-faceted scaffolding tool. However, realising this potential requires not only careful instructional design but also consideration of the practical and ethical issues of transparency and reliability in its educational deployment (Yan et al. 2023).

Building upon the reviewed literature on scaffolding, pre-reading activities, and the emerging role of AI in EFL contexts, it is essential to ground the present study in a coherent theoretical model. Such a framework not only provides conceptual clarity but also ensures that the design of the AI-supported pre-reading scaffolding intervention is systematically aligned with established learning principles. The following section outlines the integrated theoretical perspectives that underpin the study and illustrates how they inform both the intervention design and the research questions.

2.3 | Integrated Sociocultural, Scaffolding, and Motivational Perspectives for AI-Supported Pre-Reading Scaffolding

The design and analysis of this study are guided by an integrated framework that synthesizes three complementary perspectives:

Sociocultural Theory (SCT), scaffolding theory, and Self-Determination Theory (SDT) of motivation. SCT, particularly the construct of ZPD, frames learning as a socially mediated process where learners advance through guided participation. Scaffolding theory operationalizes this principle, detailing how structured, contingent support can be gradually withdrawn to foster learner autonomy (Wood et al. 1976). Complementing these cognitive-social models, Self-Determination Theory extends the framework into the affective domain. SDT posits that sustained engagement in a task is nurtured by fulfilling three basic psychological needs: autonomy (a sense of choice), competence (a sense of mastery), and relatedness (a sense of connection) (Deci and Ryan 2012; Ryan and Deci 2020).

This synthesis posits that effective scaffolding is not merely a cognitive aid but also an affective catalyst; it must simultaneously mediate comprehension and foster positive learning dispositions. Applied to the present intervention, this theoretical framework informs the design of the AI-integrated pre-reading platform in three interdependent ways. First, SCT and scaffolding theory underpin the sequencing of AI-generated prompts, ensuring that support is contingent, adaptive, and progressively withdrawn as learners gain proficiency. Second, principles from SDT guide the platform's interactive design to support learners' needs for autonomy and competence, thereby reducing affective barriers such as anxiety or disengagement. Finally, the combined framework shapes the formulation of the research questions, which examine not only the cognitive outcomes of reading comprehension but also the affective outcomes of motivation and attitudes. This dual focus reflects the premise that effective pre-reading scaffolding must engage both the mind and the disposition of the learner.

3 | Methods

3.1 | Research Designs and Study Context

This study adopted a quasi-experimental pretest-posttest control group design to examine the effects of AI-supported pre-reading scaffolding on students' English reading comprehension, motivation, and attitudes. The research was conducted within a compulsory undergraduate course, English Reading and Communication, at a university in southern Taiwan. The course, designed for non-English majors, aimed to enhance students' reading comprehension, vocabulary acquisition, and critical thinking skills through interactive, technology-enhanced instruction. The primary textbook was Reading Explorer 3 (Douglas and Bohlke 2019), supplemented by instructor-curated online resources. The course consisted of one 100-min class per week over a 16-week semester. The experimental intervention spanned eight consecutive weeks of the semester. During each weekly session in this period, the pre-reading scaffolding intervention was administered, followed by common while-reading and post-reading activities for all participants.

3.2 | Participants

The participants were 66 undergraduate students recruited from two intact classes of a compulsory English course. Utilising two

classes taught by the same instructor was intended to minimise potential teacher-related variability. The sample comprised an equal number of male and female students ($n = 33$ each), with ages ranging from 19 to 22 years ($M = 20.4$, $SD = 0.8$). They represented a variety of academic majors, including Business Administration, Information Management, Mechanical Engineering, and Civil Engineering. According to a university placement test aligned with the Common European Framework of Reference for Languages (CEFR), their English proficiency levels ranged from B1 to B2. All participants had received at least 6 years of formal English instruction prior to university. The two intact classes were assigned to the experimental and control groups, and pre-test data confirmed that the groups possessed comparable demographic and proficiency profiles at the outset of the study.

All procedures were conducted in accordance with institutional ethical guidelines. At the beginning of the semester, the course instructor fully informed all students of the study's objectives, procedures, and the voluntary nature of their participation. It was emphasised that they had the right to withdraw at any time without any penalty to their course grades. Subsequently, written informed consent was obtained from every participant before the commencement of the study.

3.3 | Experimental Procedure

The study was conducted over a 16-week semester, with the experimental intervention implemented during five class sessions in weeks 4, 6, 8, 11, and 13 (see Figure 1 for a visual timeline). This schedule was deliberately paced to align with the course curriculum and pedagogical goals. The initial session in week

4 followed foundational instruction on reading strategies, while the sessions in weeks 6 and 8 coincided with the introduction of more complex texts requiring greater support. The final two sessions after the midterm break (weeks 11 and 13) were designed to reinforce learning and sustain motivation throughout the latter half of the course.

Each intervention session followed a consistent three-stage instructional flow: pre-reading, while-reading, and post-reading. The experimental manipulation occurred during the pre-reading stage. Students in the experimental group individually interacted with a ChatGPT-powered learning platform that guided them through personalised scaffolding tasks, such as activating prior knowledge, previewing vocabulary, and generating predictions. In the control group, students received equivalent pre-reading scaffolding delivered by the instructor. These teacher-led activities, designed to mirror the objectives of the AI tasks, included whole-class discussions, PowerPoint-based vocabulary previews, and collective predictive questioning. Following their respective pre-reading activities, both groups engaged in the same while-reading comprehension tasks and participated in a common post-reading reflective discussion.

To ensure instructional consistency and fidelity, several measures were implemented. All sessions for both groups were conducted by the same experienced instructor, who adhered strictly to a detailed lesson plan for each condition (see Appendix A). This protocol ensured that while the delivery method of the pre-reading scaffolding differed, the core learning objectives, content coverage, and time allocation remained equivalent across the two groups, thereby minimising potential teacher-related variability.

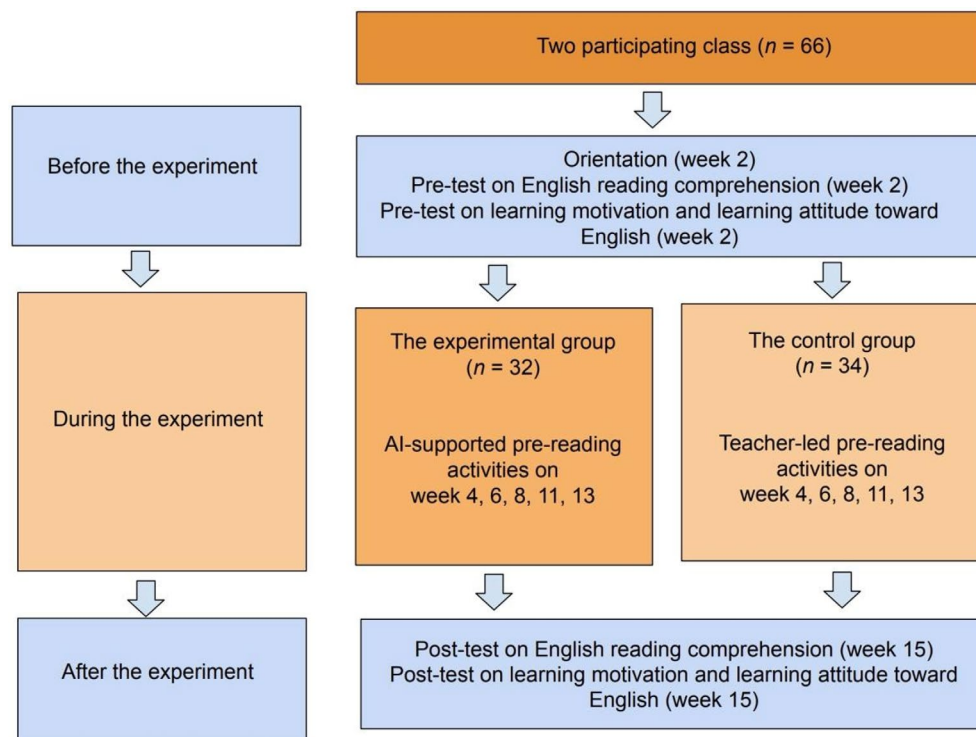


FIGURE 1 | The research procedure.

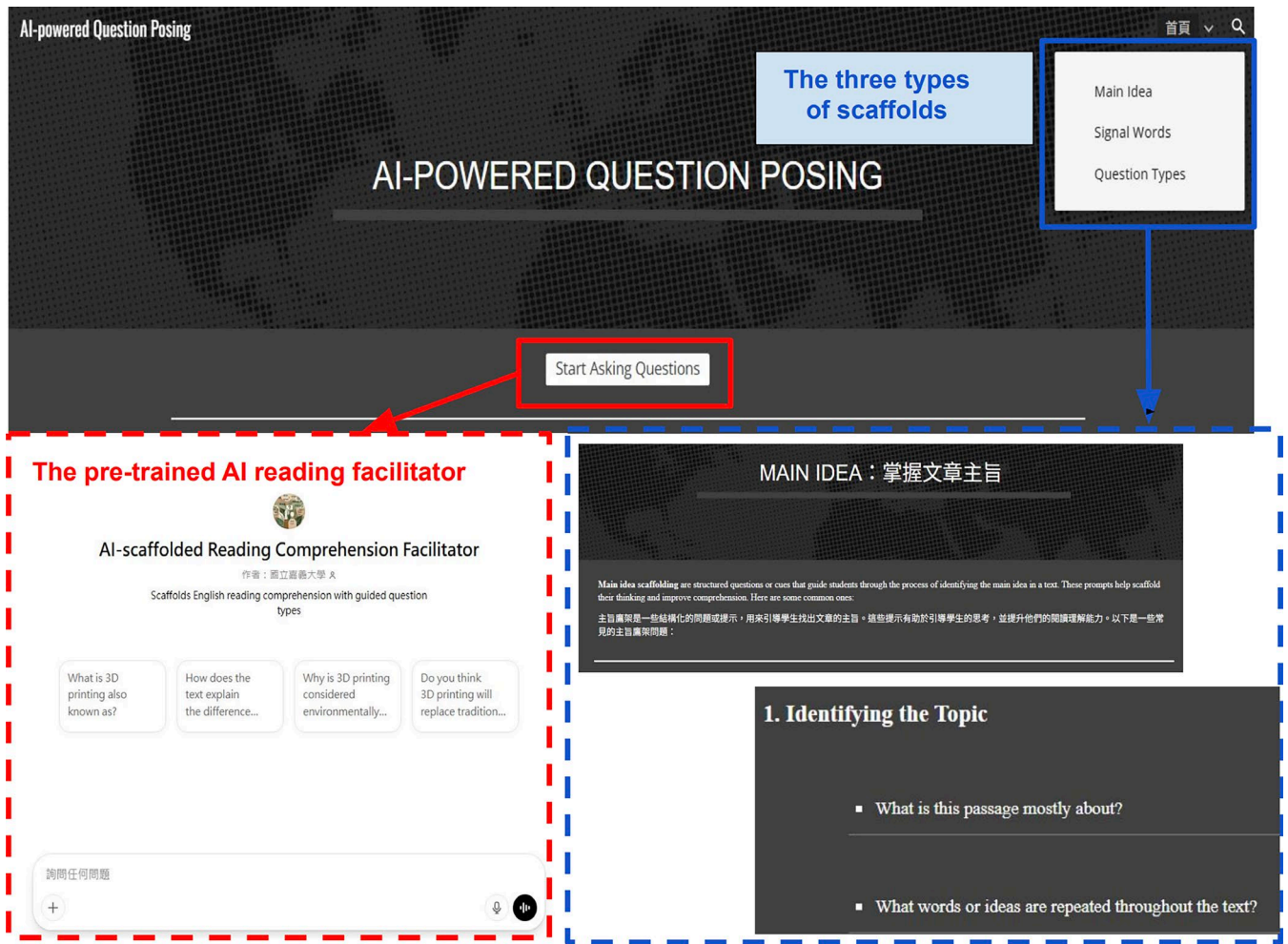


FIGURE 2 | The interface of artificial intelligence-powered question posing (AIPQP) system.

3.4 | AI-Powered Question Posing System and the Design of AI-Supported Pre-Reading Scaffolding

For the experimental condition, we developed an online AI-powered Question Posing (AIPQP) system that integrated a pre-trained version of ChatGPT (see Figure 2). Grounded in procedural scaffolding theory, the system was designed to guide students through structured metacognitive routines during the pre-reading phase. Specifically, the interface facilitated three key scaffolding strategies: (1) identifying main ideas, (2) recognising signal words, and (3) generating and responding to layered comprehension questions. These components were intended to activate learners' prior knowledge, guide content prediction, and encourage reflective reasoning before engaging with the main text.

The AIPQP system is conceptually parallel to the ChatPRCS system proposed by Wang et al. (2024), which validated the use of ChatGPT for generating personalised reading questions. However, our system introduces a key pedagogical distinction by shifting the focus from AI-generated question-answering to student-initiated question-posing. Whereas the ChatPRCS model positions the AI as the question generator, the AIPQP system empowers learners to actively construct their own inquiries. This shift from a receptive to a constructive learning role is central to our design, aiming to foster deeper cognitive

processing and metacognitive awareness in alignment with the “question generation effect.” This approach was operationalised through a detailed prompt engineering strategy (see an example in Figure 2). Prompts were engineered to be open-ended yet structured, while the adaptive nature of the LLM allowed the system to provide tailored follow-up questions and elaborative feedback based on each student's unique inquiries, ensuring a responsive and personalised scaffolding experience.

To facilitate this process, students in the experimental group were trained to formulate relevant, text-based questions using a set of standardised templates. Within the system, ChatGPT was engineered to provide contextually appropriate, elaborative responses, such as paraphrased summaries and inferential clarifications, rather than direct answers. This design encouraged multi-turn dialogic engagement, prompting students to refine their inquiries and deepen their critical thinking. Figure 3 provides a representative example of this implementation, illustrating the prompt engineering strategy and the resulting dialogic guidance. As shown, the structured prompts were designed to scaffold student inquiry, promote text-based reasoning, and avoid direct answer provision, thereby fostering independent comprehension.

Figure 3 also illustrates the structured prompt design and interaction flow of the AI-supported pre-reading scaffolding. Engineered prompts guide students through multiple levels of



FIGURE 3 | Structured prompt design and ChatGPT interaction flow in AI-supported pre-reading scaffolding.

comprehension and metacognitive reflection. The system provides tailored, elaborative feedback while withholding direct answers, an approach designed to enhance both cognitive and affective engagement in English reading tasks.

3.5 | Measurements

Regarding the affective outcomes, this study assessed learners' general motivation and attitudes towards English learning, rather than focusing solely on reading-specific constructs. This decision is grounded in empirical evidence demonstrating strong links between reading-specific affect and broader language learning orientations. For instance, research indicates that reading motivation and self-efficacy are structurally linked to overall comprehension (Anggia and Habók 2025), and that enjoyment of reading significantly predicts general language achievement (Li and Gan 2022). These findings support the rationale that positive experiences within a specific skill domain, such as reading, can transfer to and influence a learner's overall affective stance towards the language. Therefore, assessing general motivation and attitudes was chosen to provide a more holistic understanding of the intervention's potential impact.

3.5.1 | English Reading Comprehension Tests

The English reading comprehension test, administered as both a pretest and posttest, consisted of two passages, each accompanied by 10 multiple-choice questions (20 items total). The items were designed to assess four distinct levels of comprehension: literal, inferential, evaluative, and strategic. The reading passages were selected from materials aligned with the B2 level of the CEFR. The suitability of the texts was confirmed through evaluation by two experts with over 10 years of tertiary-level English teaching experience and cross-checked using the Flesch–Kincaid readability index. Following a pilot administration, items were revised or replaced if their difficulty index fell outside the acceptable range of 0.30–0.80. A detailed test specification table is provided in Appendix B. The maximum score for the test was 20 points, and the time limit for completion was 40 min.

3.5.2 | Learning Motivation

Learning motivation was assessed using a 14-item questionnaire grounded in the expectancy-value model (Pintrich and Schunk 1996) and subsequently adapted by Hong (2002) for

EFL learners. The instrument comprises two subscales: Work Value (6 items) and Self-Efficacy and Expectancy (8 items). Participants responded to all items on a six-point Likert scale, ranging from 1 (strongly disagree) to 6 (strongly agree). The questionnaire demonstrated high internal consistency in the present study, with Cronbach's alpha coefficients of 0.92 for the Work Value subscale, 0.94 for the Self-Efficacy and Expectancy subscale, and 0.94 for the overall instrument. The questionnaire was administered at both pre- and post-intervention phases to measure changes over time.

3.5.3 | Learning Attitude Towards English

Learners' attitudes towards English were assessed using an instrument adapted from Hsieh (2017) for university-level students. The final 29-item scale comprises three subscales: Self-Confidence (11 items), Enjoyment (9 items), and Perceived Anxiety (9 items). All items were rated on a four-point Likert scale ranging from 1 (strongly disagree) to 4 (strongly agree). The instrument underwent a pilot validation study with a separate sample ($n = 166$), during which an initial pool of 32 items was refined to the final 29 after three items with low factor loadings were removed. The resulting scale demonstrated excellent internal reliability in the present study (Cronbach's $\alpha = 0.94$) and was administered in both the pre- and post-test phases.

3.5.4 | The Post-Activity Open-Ended Survey

To gather in-depth qualitative insights, a post-activity open-ended survey was administered to participants following the intervention. The survey consisted of four questions designed to capture a range of learner experiences. The first question probed the perceived benefits of the AI integration on aspects such as comprehension, strategy use, and engagement. The second explored its influence on learning motivation, including both intrinsic and extrinsic factors. The third investigated any changes in learners' attitudes towards English as a result of the AI-mediated scaffolding. The final question invited participants to describe any challenges they encountered, whether technological, cognitive, or affective. This qualitative instrument was designed to provide rich, contextualised evidence to complement the study's quantitative findings.

3.6 | Data Analysis

Quantitative data were analysed using Analysis of Covariance (ANCOVA) to compare the post-test scores of the experimental and control groups while controlling for pre-test scores as a covariate. ANCOVA was selected over an independent samples t-test because it statistically adjusts for baseline differences between the intact groups, providing a more accurate estimation of the intervention's effects. This approach is particularly suitable for the study's quasi-experimental design where random assignment was not feasible. Prior to the main analysis, all underlying assumptions for ANCOVA were verified: homogeneity of regression slopes, normality of residuals (assessed via the Shapiro-Wilk test and Q-Q plots), and homoscedasticity (evaluated with Levene's test). All assumption tests were

non-significant ($p > 0.05$), confirming the appropriateness of ANCOVA.

Qualitative data from the open-ended survey were analysed using thematic analysis, following the procedures outlined by Braun and Clarke (2006). The analysis was facilitated by NVivo 14 software to ensure systematic data management and an auditable analytic trail. The process involved an inductive approach, beginning with open coding of the raw responses, followed by the categorization of these codes and the development of overarching themes. For instance, codes such as enhanced vocabulary learning and increased engagement were synthesised into the theme Perceived Learning Benefits, while codes like technical issues and unfamiliarity with AI formed the theme Challenges Encountered.

4 | Results

4.1 | English Reading Comprehension Performance

Table 1 presented the descriptive statistics of students' English reading comprehension scores for the teacher-led and AI-supported scaffolding conditions. Before the intervention, the teacher-led group had a pretest mean score of 73.38 (SD = 13.58), while the AI-supported group had a slightly higher mean of 78.90 (SD = 11.34). The pre-reading test results indicated that the AI-supported group scored higher than the teacher-led group. An independent-samples t-test was conducted to examine whether this difference was statistically significant. The results revealed no significant difference between the two groups at the pretest, $t(64) = 1.97$, $p = 0.053$, indicating that the groups were statistically equivalent in their initial reading comprehension performance. Thus, this supports the appropriateness of using ANCOVA to control for pretest scores when examining post-intervention differences.

After the intervention, the posttest mean scores were 74.85 (SD = 12.02) for the teacher-led group and 82.34 (SD = 11.07) for the AI-supported group. After adjusting for pretest scores using ANCOVA, the adjusted posttest means were 75.33 for the teacher-led group and 81.82 for the AI-supported group, indicating a notable advantage in favour of the AI-supported pre-reading scaffolding.

To examine the effect of instructional method on students' English reading comprehension performance, an analysis of covariance (ANCOVA) was conducted using pretest scores as

TABLE 1 | Descriptive statistics and adjusted means for English reading comprehension scores.

Groups	M	SD	M	SD	Adjusted
					M
Teacher-led ($n = 34$)	73.38	13.58	74.85	12.02	75.33
AI- supported ($n = 32$)	78.9	11.34	82.34	11.07	81.82

a covariate. Box's M test indicated no violation of the assumption of homogeneity of variance–covariance matrices, Box's $M=13.06$, $p=1.00$. The results of ANCOVA (see Table 2) revealed a statistically significant effect of instructional method on posttest reading comprehension scores after adjusting for pretest performance, $F(1, 63)=5.06$, $p=0.02$, partial $\eta^2=0.07$, indicating a moderate effect size (partial $\eta^2=0.07$). According to Cohen (2013), this corresponds to a medium effect, suggesting that the scaffolding delivery mode accounted for a meaningful portion of the variance in reading comprehension scores.

4.2 | English Learning Motivation

Table 3 displays the descriptive statistics for students' learning motivation scores in the teacher-led and AI-supported scaffolding groups. Before the intervention, the teacher-led group reported a mean motivation score of 17.26 (SD=5.53), while the AI-supported group had a higher mean of 21.15 (SD=7.34). The AI-supported group had a higher motivation pretest mean compared to the teacher-led group. An independent samples t-test indicated that this pretest difference was not statistically significant, $t(64)=1.86$, $p=0.067$, suggesting that the baseline motivation levels between the two groups were comparable prior to the intervention.

After the intervention, the posttest scores increased in both groups, with the teacher-led group reaching a mean of 22.97

TABLE 2 | ANCOVA summary for English reading comprehension performance.

Variable	SS	df	MS	F	p	Partial η^2
Group	661.43	1	661.43	5.06	0.02*	0.07
Pre-test	330.27	1	330.27	2.52	0.11	0.03
Error	8243.21	63	130.06			

Note: * $p < 0.05$.

TABLE 3 | Descriptive statistics and adjusted means for learning motivation scores.

Groups	Pre-test		Post-test		Adjusted M
	M	SD	M	SD	
Teacher-led ($n=34$)	17.26	5.53	22.97	5.41	23.86
AI-supported ($n=32$)	21.15	7.34	27.18	4.16	26.23

TABLE 4 | ANCOVA summary for learning motivation.

Variable	SS	df	MS	F	p	Partial η^2
Group	84.6	1	84.06	5.9	0.01*	0.08
Pre-test	606.29	1	606.29	42.46	0.001	0.4
Error	899.55	63	14.27			

Note: * $p < 0.05$.

(SD=5.41), and the AI-supported group attaining a mean of 27.18 (SD=4.16). After adjusting for pretest differences using ANCOVA, the adjusted posttest means were 23.86 for the teacher-led group and 26.23 for the AI-supported group, indicating a sustained motivational advantage for those receiving AI-supported pre-reading scaffolding.

Box's M test indicated no significant violation of the homogeneity of covariance assumption, Box's $M=8.06$, $F(2, 81964.75)=2.68$, $p=0.054$. Thus, the assumption was considered acceptable for ANCOVA.

The ANCOVA (see Table 4) revealed a significant main effect of instructional method on posttest learning motivation scores after controlling for pretest performance, $F(1, 63)=5.90$, $p=0.01$, partial $\eta^2=0.08$, indicating a moderate effect size (partial $\eta^2=0.08$). Based on Cohen's (2013) guidelines, this represents a medium effect, highlighting the practical significance of the intervention on learners' motivational outcomes.

4.3 | Learning Attitude Towards English

Table 5 presents the descriptive statistics for subject-related attitude scores across the teacher-led and AI-supported pre-reading scaffolding groups. Prior to the intervention, both groups showed similar levels of subject-related attitude, with the teacher-led group scoring a mean of 47.96 (SD=6.99) and the AI-supported group scoring 48.68 (SD=5.95). After the intervention, the posttest mean score of the AI-supported group increased to 51.93 (SD=6.80), whereas the teacher-led group slightly decreased to 45.90 (SD=5.60). An independent samples t-test indicated that this initial difference was not statistically significant, $t(64)=-0.45$, $p=0.65$, confirming baseline equivalence between the groups.

After adjusting for pretest scores using ANCOVA, the adjusted means were 45.94 for the teacher-led group and 51.89 for the AI-supported group, suggesting a favourable impact of AI-supported scaffolding on learners' learning attitude towards English.

Box's M test indicated no significant violation of the homogeneity of variance–covariance assumption, Box's $M=2.33$, $F=0.78$, $p=0.511$. Therefore, the assumption was deemed tenable for conducting ANCOVA.

The ANCOVA (see Table 6) revealed a statistically significant main effect of instructional method on posttest subject-related attitude, $F(1, 63)=14.77$, $p < 0.001$, partial $\eta^2=0.19$, indicating a large effect size. According to Cohen (2013), this effect is considered

large, suggesting a substantial impact of AI-supported scaffolding on students' affective orientation towards English learning.

4.4 | Participants' Reflective Feedback

Qualitative analysis of students' reflective responses revealed a major perception that AI-supported pre-reading scaffolding enhanced their reading comprehension (see Table 7). Among the 32 students in the AI group, 28 explicitly stated that AI tools helped them better understand the reading materials, particularly in relation to vocabulary meaning, textual structure, and identifying main ideas.

4.4.1 | Theme 1: Enhanced Reading Comprehension

Most students ($n=28$) reported that interacting with AI helped clarify textual content and improve their understanding of word meanings and main ideas. A few participants further indicated that AI-generated explanations helped them locate supporting textual evidence and better understand why certain answers were correct.

- “It helped me understand the vocabulary and meaning of the article more clearly” (S1).
- “Sometimes AI provided deeper analysis of the answer, so I could better locate the supporting part of the text” (S2).
- “When I didn't have the patience to finish reading, I asked AI to summarize the key points” (S3).
- “When I encountered something I didn't understand, it directed my attention to the key points of the article” (S5).

4.4.2 | Theme 2: Increased Learning Motivation

A total of 15 students explicitly reported that interacting with the AI system increased their motivation to read in English. Several students noted that AI support helped reduce the

TABLE 5 | Descriptive statistics and adjusted means for learning attitude towards English scores.

Groups	<i>M</i>	<i>SD</i>	<i>M</i>	<i>SD</i>	Adjusted <i>M</i>
Teacher-led ($n=34$)	47.96	6.99	45.9	5.6	45.94
AI-supported ($n=32$)	48.68	5.95	51.93	6.8	51.89

TABLE 6 | ANCOVA summary for learning attitude towards English.

Variable	SS	df	MS	<i>F</i>	<i>p</i>	Partial η^2
Group	572.48	1	572.48	14.77	0.001**	0.19
Pre-test	35.62	1	35.62	0.92	0.034	0.01
Error	2402.98	63	38.76			

Note: ** $p < 0.001$.

frustration associated with difficult texts, enabling them to better understand the material and feel more competent. Others mentioned that the novelty and usefulness of the AI tool made reading more enjoyable and less intimidating, thus increasing their willingness to engage with the task. The motivational impact appeared to stem from AI's ability to provide immediate support, clarify difficult segments, and foster a sense of progress and autonomy.

- “I could understand English texts more easily, so I felt more motivated” (S1).
- “I used to feel frustrated because of my limited English ability, but AI helped me know where to begin” (S21).
- “It was fun. I didn't feel so resistant to English reading when AI helped me analyse the passage” (S16).

4.4.3 | Theme 3: Improved Attitude Towards English Learning

Ten students specifically noted that the use of AI increased their emotional engagement and positive disposition towards English learning. They described the AI-supported activity as convenient, low-pressure, and even enjoyable, which led to increased

TABLE 7 | Summary of thematic analysis on student perceptions of AI-supported pre-reading scaffolding.

Theme	Sub-theme	Frequency of mention
Enhanced English reading comprehension	Vocabulary and passage understanding	30/32
	Identification of main ideas	7/32
Enhanced learning motivation	Improved learning motivation	11/32
	Perceived as more engaging	6/32
Increased learning attitude towards English	More willingness and less anxiety	27/32
	Perceived ease and enjoyment	5/32
Challenges encountered when using AI	Unfamiliarity with AI usage	4/32
	Inaccuracy or off-topic responses	4/32

interest and willingness to learn. For these students, the integration of AI into the classroom shifted their perception of English from being difficult and stressful to approachable and engaging.

- “I’ve made a lot of progress. Learning English feels a bit easier, and I don’t feel as much pressure as before” (S2).
- “My feeling is that after using AI, I felt more confident when answering English reading questions and was less afraid of making mistakes” (S19).
- “After integrating the tool (AI), I felt that reading became easier, and I no longer felt so resistant to English” (S32).

4.4.4 | Theme 4: Challenges Encountered in Using AI

A subset of students reported experiencing challenges while interacting with the AI system. These included difficulty understanding overly complex or vague responses, instances where the AI provided off-topic or incomplete answers, and the need to ask follow-up questions for clarification. Although such challenges were not frequently mentioned, they highlight limitations in the AI’s responsiveness and transparency. Despite these issues, students generally acknowledged the overall usefulness of AI, suggesting that the occasional misunderstandings did not outweigh the benefits.

- “Sometimes it responded to irrelevant questions, though most of it was still helpful” (S2).
- “Sometimes the AI might provide incorrect information” (S4).
- “Yes, sometimes the AI gave off-topic answers” (S8).

5 | Discussion

5.1 | English Reading Comprehension Performance

The findings demonstrate that generative AI-supported pre-reading scaffolding enhanced university EFL learners’ reading comprehension. After controlling for pre-existing differences, students who engaged with the AI-supported scaffolding outperformed their peers in the teacher-led condition. The moderate effect size (partial $\eta^2 = 0.07$) indicates that the intervention had a practically meaningful impact, accounting for 7% of the variance in post-test scores. This suggests that the AI-supported approach offers not just a statistically significant, but a tangible pedagogical advantage, highlighting the potential of Gen-AI to serve as a cognitively responsive scaffolding agent.

This result aligns with established research emphasising the critical role of pre-reading activities in facilitating comprehension (Ajideh 2006; Alemi and Ebadi 2010; Wijaya 2023). It also extends recent work showing that interactive digital interfaces can improve textual processing (Bahari et al. 2024). While activating prior knowledge and previewing content are known to be effective, the key contribution of the present study lies in its focus on the delivery mode. Instead of relying on synchronous, whole-class teacher instruction, the scaffolding was embedded

within a structured AI interface, providing learners with immediate, individualised, and repeatable support.

The cognitive advantage of the AI-based system may stem from its ability to support the complex mental processes essential for reading, such as inference generation and the integration of background knowledge (Rumelhart and McClelland 2017). The structured question types on the AIPQP system likely prompted learners to engage in these interpretive processes more actively, a notion consistent with Swain’s (2005) Output Hypothesis and Long’s (1996) Interaction Hypothesis. This interpretation is corroborated by our qualitative findings; a majority of participants in the AI group (28 out of 32) reported that the scaffolding helped them clarify main ideas and locate key information. One student noted, “Sometimes AI provided deeper analysis of the answer, so I could better locate the supporting part of the text.” This suggests learners engaged in self-directed meaning-making congruent with deep reading processes.

However, these positive findings must be interpreted with caution. The literature notes that while generative AI provides personalised support, its output can be inaccurate or pedagogically inappropriate (Bang et al. 2023). Furthermore, there is a risk of learners developing an over-reliance on AI, which could hinder their development of independent metacognitive strategies (Hariri 2023). Therefore, the effectiveness of AI scaffolding is contingent not just on the technology itself, but on its integration into a pedagogical framework that promotes critical thinking and learner autonomy.

Furthermore, the comparatively lower gains in the teacher-led group warrant a nuanced interpretation. Rather than an indictment of teacher-led instruction per se, this result may reflect the inherent constraints of a typical university classroom. The social dynamics of a group setting can elevate learners’ affective filters, making them hesitant to ask fundamental questions or express uncertainty for fear of judgment. In contrast, the AI provided a private, non-evaluative space for inquiry. Thus, the advantage of the AI group may stem as much from its capacity to create a psychologically safe learning environment as from its technological capabilities.

In summary, the observed enhancement in reading comprehension underscores the efficacy of using generative AI to deliver pre-reading support. The AI-integrated approach not only fulfills the core functions of traditional pre-reading activities but also offers distinct advantages in flexibility, personalization, and creating a psychologically safe learning environment. These findings position Gen-AI systems as valuable tools for facilitating reading comprehension, particularly in resource-constrained EFL university contexts.

5.2 | Learning Motivation

The study also revealed that AI-supported scaffolding significantly enhanced students’ learning motivation. The system’s capacity for personalised feedback and multi-turn dialogue created an environment conducive to meaning negotiation and output modification, aligning with interactionist SLA theories (Long 1996; Swain 2005). This result empirically supports

recent scholarship suggesting that thoughtfully integrated AI systems can positively influence the affective dimensions of language learning (Liu et al. 2024; Waluyo and Kusumastuti 2024).

From a motivational psychology perspective, the AI system likely nurtured intrinsic motivation by supporting learners' basic psychological needs for autonomy and competence. Unlike conventional whole-class instruction, the AI interface offered a private, self-paced environment where learners had greater control over their learning process. This autonomy-supportive context can reduce affective barriers such as anxiety, encourage ownership of the task, and enhance feelings of competence, all of which are key predictors of sustained motivation (Ryan and Deci 2020).

These quantitative patterns were echoed in students' reflective responses. Nearly half of the participants (15 out of 32) in the AI group explicitly stated that the system increased their motivation to read in English, often linking it to reduced frustration and a greater sense of manageability. One student commented, "I could understand English texts more easily, so I felt more motivated," while another reflected, "Using AI made learning smoother, and I wanted to try more questions." Such feedback highlights the AI's role in lowering anxiety and enhancing perceived competence, central components of self-determined motivation.

However, these strong affective outcomes must be interpreted with consideration for the novelty effect. The initial excitement of using a sophisticated AI tool in a formal course may have temporarily inflated reported motivation and attitudes, a common factor in short-term technology interventions. Therefore, the observed moderate effect size for motivation (partial $\eta^2=0.08$) and the large effect size for attitudes (partial $\eta^2=0.19$), while practically meaningful, should be viewed with this caution. A portion of these gains may be attributable to this initial enthusiasm rather than a stable, long-term change in disposition. Future longitudinal research is essential to track whether these affective gains are sustained after the novelty has subsided.

Overall, the significant motivational gains observed in the AI condition highlight the capacity of generative AI to support not only cognitive processes but also to foster more positive learner dispositions. By functioning as both a cognitive tool and an affective support, Gen-AI demonstrates its considerable potential for designing more learner-centered, engaging, and motivating digital language instruction.

5.3 | Learning Attitude Towards English

Finally, the intervention had a substantial positive impact on students' attitudes towards English learning, a finding supported by a large effect size (partial $\eta^2=0.19$). This indicates that the AI-supported scaffolding had not only a statistically significant but also a transformative practical impact on learners' dispositions. Such a strong affective shift is critical in EFL university contexts, as student attitudes are known predictors of long-term engagement and academic achievement (Huang and Hong 2016; Li and Gan 2022). This suggests that well-designed Gen-AI integration can be a powerful tool for reshaping students' emotional engagement with a language.

The AI-supported scaffolding likely fostered these positive attitudes by creating a psychologically safe and non-judgmental learning environment. The private, text-based nature of the interaction removed the social pressures and performance anxiety often associated with real-time, face-to-face classroom exchanges. This may have helped lower the affective filter (Krashen 1982), allowing students to engage with English materials with greater confidence and openness.

Student reflections provide further evidence for this interpretation. A number of participants explicitly stated that using the AI tool made them feel more comfortable and positive about learning English. For instance, one student noted, "After integrating the tool (AI), I felt that reading became easier, and I no longer felt so resistant to English." Another reflected, "Using AI made English class feel more relaxed and enjoyable." These accounts directly link the AI tool to reduced resistance and increased comfort, suggesting the system was effective in reshaping learners' emotional orientation towards the subject.

However, it is important to critically reflect on the overwhelmingly positive nature of this feedback and consider potential response biases. As previously noted, the enthusiasm surrounding the novelty of using a cutting-edge tool may have led to more favourable perceptions than would be found over a longer period. Additionally, a social desirability bias may have influenced students to provide feedback they perceived as favourable to the researcher. These factors might partly explain why few students reported challenges. Therefore, future research should complement self-report surveys with more indirect methods, such as observational data or think-aloud protocols, to capture a more unvarnished picture of the learner experience.

6 | Conclusion and Pedagogical Implications

This study demonstrated that AI-supported pre-reading scaffolding, grounded in structured question posing, effectively improved university EFL learners' reading comprehension, motivation, and attitudes. The integration of generative AI served not only as a cognitive tool but also as an affective support mechanism by offering individualised, low-pressure, and feedback-rich interactions.

These findings offer several actionable pedagogical implications for higher education. First, educators can integrate AI scaffolding tools into curricula to provide personalised support, especially in large or resource-constrained classrooms. Second, teacher training programs should incorporate modules on prompt engineering and the pedagogical design of AI-driven dialogues to ensure alignment with instructional goals. Third, institutions can leverage these tools in blended or asynchronous formats to enhance learner autonomy, a benefit supported by recent research showing that frequent AI chatbot interaction fosters autonomy in virtual contexts (Xie et al. 2024). Finally, curriculum designers should ensure that AI-supported activities are coherently aligned with broader language learning outcomes and digital literacy frameworks.

Theoretically, this study extends scaffolding theory into AI-mediated environments, providing evidence that generative AI

can function as a dynamic scaffolding agent. It also contributes to Self-Determination Theory by identifying AI interaction as a novel means of supporting learners' needs for autonomy and competence. Building on these findings, future research should explore several key areas. Investigations are needed into adaptive prompt sequencing based on learner performance, the qualitative dimensions of learner–AI interaction, and the longitudinal impacts of sustained AI use on language development and metacognitive skills.

6.1 | Limitations and Suggestions for Future Studies

While this study provides valuable insights, several limitations should be acknowledged as they offer important directions for future research. First, the quasi-experimental design and the short-term nature of the intervention constrain internal validity and the generalizability of the findings. Future research should employ randomized controlled trials (RCTs) and longitudinal designs to establish causality more firmly and assess the long-term sustainability of the observed effects. Additionally, the study's focus on university-level EFL learners in a single institutional context limits external validity. Subsequent studies should examine diverse learner populations across different proficiency levels, age groups, and educational systems.

Second, the design of the scaffolding itself warrants further investigation. The present study did not disaggregate the effects of specific question types (e.g., literal, inferential), so future work could analyse which scaffolding elements are most effective for different comprehension sub-skills. Furthermore, while standardising the prompts ensured consistency, it also constrained a key affordance of generative AI: dialogic freedom. Our findings demonstrate AI's effectiveness in a controlled interaction, but future research should compare such structured models with more open-ended, student-led dialogic approaches to better understand the trade-offs between pedagogical structure and learner autonomy.

Finally, future inquiries should move beyond the dichotomy of AI-versus-teacher scaffolding to explore integrated or hybrid models that leverage the respective strengths of both. From a technical standpoint, the system used in this study lacked built-in mechanisms to proactively prevent AI hallucinations. While our structured prompts mitigated this risk, future designs should incorporate automated reliability checks and content filters to enhance the safety and trustworthiness of such tools for classroom use.

Author Contributions

Chih-Chung Lin: conceptualization, investigation, writing – original draft, methodology, validation, visualization, writing – review and editing, formal analysis, project administration, supervision, resources, funding acquisition, software, data curation. **Tzu-Hsuan Lin:** software, data curation, writing – review and editing. **Chi-Kay Tang:** software, data curation, writing – review and editing.

Conflicts of Interest

The authors declare no conflicts of interest.

Data Availability Statement

The data that support the findings of this study are available on request from the corresponding author. The data are not publicly available due to privacy or ethical restrictions.

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Appendix A

Lesson Plan and Instructional Flow

Step 1 (10 min) Vocabulary Preview and Semantic Mapping

Students are introduced to 8–10 key vocabulary items and academic phrases selected from the target text. For the experimental group, the AI platform provides definitions, example sentences, and prompts students to generate their own sentences in context. For the control group, the instructor presents definitions and examples on slides, followed by a brief class discussion. Both groups collaboratively create a semantic

map linking key terms to the text's overarching theme, thereby activating relevant schema.

Step 2 (15 min): Prior Knowledge Activation

In the experimental group, students interact with ChatGPT via the AI-integrated learning platform. The AI poses tailored, open-ended questions (e.g., “What do you already know about renewable energy sources?”) and provides adaptive follow-up prompts based on each student’s response, fostering elaboration and clarification. In the control group, the teacher facilitates a whole-class discussion on the topic, eliciting students’ prior experiences and knowledge without AI mediation.

Step 3 (20 min): Predictive Questioning

Students are guided to examine text headings, subheadings, and visuals to predict content. In the experimental group, ChatGPT generates personalized predictive questions and offers feedback on students’ initial hypotheses, encouraging revision and refinement. In the control group, the teacher leads the predictive questioning process collectively. Both

groups record predictions in their reading journals for later comparison with the actual text content.

Step 4 (30 min): While-Reading Comprehension Tasks

Students read the assigned passage individually. Comprehension tasks include multiple-choice and short-answer questions targeting literal, inferential, and evaluative comprehension. In the experimental group, AI provides immediate, item-specific feedback and offers hints or simplified rewording when a student struggles. In the control group, the teacher reviews answers collectively after completion.

Step 5 (15 min): Post-Reading Reflection and Group Sharing

Students compare their initial predictions to the actual content and reflect on discrepancies. In the experimental group, AI prompts individual reflective responses and suggests ways to improve future prediction accuracy. In both groups, students share their reflections in small groups, followed by a class-wide synthesis discussion facilitated by the teacher.

Appendix B

Test Specification Table and Sample Items

Passage	No. of items	Comprehension type	Item no.	Example question stem	Difficulty index
1. AI	10	Literal	1–3	According to the text, what is the main reason for the recent surge in AI applications?	0.65
		Evaluative	4–6	Based on the text, what is a potential concern about AI adoption?	0.58
		Inferential	7–8	What is the main difference between narrow AI and general AI?	0.62
		Strategic	9–10	How does AI learning relate to natural selection?	0.48
2. Robotics	10	Literal	1–2	What is the primary purpose of the FIRST Robotics Competition?	0.70
		Evaluative	3–4	What can be concluded about the future of robotics?	0.66
		Inferential	5–7	What is the key difference between mechanical and electrical aspects?	0.59
		Strategic	8–10	How does robotics relate to disruptive innovation?	0.55